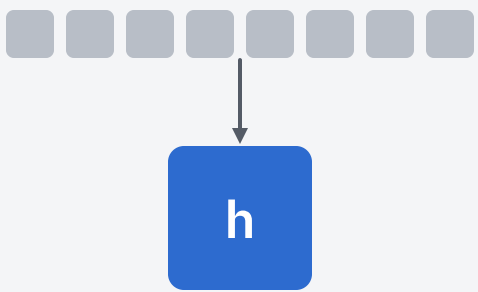
FIGURE 1

Three ways to carry the past forward

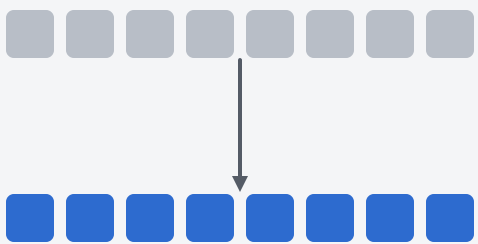
Recurrent network



one state, same rule always

+ constant memory
+ fast generation
– trains step by step
– forgets distant words

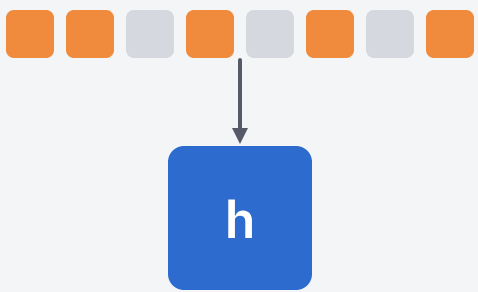
Transformer



a cache of every token

+ exact recall
+ parallel training
– memory grows per token
– cost grows with length

Mamba



one state, written by choice

+ constant memory
+ parallel training
+ chooses what to keep
– forgets what it skipped